



## Groups definitions (CH-3)

A group  $G$  is a set of elements together with an internal composition law (denoted " $\circ$ ") such that the following axioms are satisfied:

i) Neutral element:

$$\exists e \in G \mid \forall g \in G \quad g \circ e = e \circ g = g$$

ii) Inverse element:

$$\exists g_1 \in G, \exists g_2 \in G \mid g_1 \circ g_2 = g_2 \circ g_1 = e$$
$$g_2 \equiv g_1^{-1}$$

iii) Associativity:

$$\forall g_1, g_2 \in G \mid (g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$$
$$\exists g_3 \in G$$

We distinguish 3 types of groups

### ★ Finite groups

The number of elements is finite and called the order of the group

↳ ex: group of permutation:  $S_n$   
↳ order =  $n!$

### ★ Discrete infinite groups

The number of elements is infinite but countable

ex: integers with addition as composition law

### ★ Continuous infinite groups

The number of elements is infinite AND cannot be counted

All the elements depend on one or several continuous parameters

The number of these parameters defines the dimension of the group

↳ rotations

## More definitions:

### Semi-group

It is a set of elements which satisfy associativity  $\oplus$  closure but not inverse and neutral element

closure:  $g_1, g_2 \in G \mid g_1 \circ g_2 \in G$

Example:  $\mathbb{N}^*$  (natural numbers without zero) with addition

### Commutators:

The commutators of any two element  $g_1$  and  $g_2 \in G$  is the element

$$g_1 g_2 g_1^{-1} g_2^{-1} \equiv [g_1, g_2]$$

If  $g_1$  and  $g_2$  commute  $g_1 g_2 = g_2 g_1$

$$g_1 g_2 g_1^{-1} g_2^{-1} = g_2 g_1^{-1} g_1 g_2^{-1} = e$$

### ★ Abelian groups

A group  $G$  is abelian if and only if internal composition law is commutative

$$G \text{ abelian} \iff g_1 g_2 = g_2 g_1$$

ex Set  $\mathbb{Z}$  with addition

### Order of an element of a finite group:

$G$  is a finite group;  $g \in G$

the order of element  $g$  is the smallest positive integer,  $n$  such that  $g^n = e$

$$g^n = \underbrace{g \cdot g \cdot g \cdots g}_{n \text{ times}} = e$$

## Cyclic groups

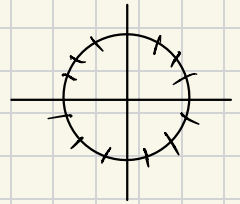
A cyclic group is the set of all the powers of an element  $g \in G$

Its order is the order of  $g$ .

A cyclic group is abelian

Notation:  $C_n$ ;  $g^n = e$

Ex planar rotations around a fixed point  $O$  with angle  $\left(\frac{2\pi}{n}\right)m$ ,  $m = 1, 2, \dots, n$



## Multiplication tables (finite groups)

It can be viewed as the ID card of a finite group

It defines the structure of the group

Its an  $(n \times n)$  array ( $n$  order of the group)

| $G$      | $g_1$     | $g_2$     | $g_3$     | $\dots$  | $g_n$     |
|----------|-----------|-----------|-----------|----------|-----------|
| $g_1$    | $g_1^2$   | $g_1 g_2$ | $g_1 g_3$ | $\dots$  | $g_1 g_n$ |
| $g_2$    | $g_2 g_1$ | $g_2^2$   | $\dots$   | $\dots$  | $\dots$   |
| $\vdots$ | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $\vdots$  |
| $\vdots$ | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $\vdots$  |
| $\vdots$ | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $\vdots$  |
| $\vdots$ | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $\vdots$  |
| $\vdots$ | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $\vdots$  |
| $g_n$    | $\vdots$  | $\vdots$  | $\vdots$  | $\vdots$ | $g_n^2$   |

## ★ Rearrangement Theorem

Let us take any element of  $G$ ,  $g$

Then composing (on the left or on the right) by  $g$ , all the elements of  $G$

We recover once and only once all the elements of  $G$ , but in a different order

→ implies that in the multiplication table in any row and in every column, each element must appear once.

## The symmetric group (or group of permutations) $S_n$

$S_n$  is the group of all permutations of an  $n$ -element set.

Let's denote  $p$  a permutation of  $n$  elements by

$$p = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ p_1 & p_2 & p_3 & \dots & p_n \end{pmatrix} = \begin{pmatrix} 2 & 5 & 1 & n & \dots & 3 \\ p_2 & p_5 & p_1 & p_n & \dots & p_3 \end{pmatrix}$$

The composition of two permutations  $p$  and  $q$

$$pq = \begin{pmatrix} 1 & 2 & \dots & n \\ p_1 & p_2 & \dots & p_n \end{pmatrix} \begin{pmatrix} 1 & 2 & \dots & n \\ q_1 & q_2 & \dots & q_n \end{pmatrix} = \begin{pmatrix} 1 & 2 & \dots & n \\ p_1 & p_2 & \dots & p_n \end{pmatrix} \begin{pmatrix} p_1 & p_2 & \dots & p_n \\ q_{p_1} & q_{p_2} & \dots & q_{p_n} \end{pmatrix} = \begin{pmatrix} 1 & 2 & \dots & n \\ q_{p_1} & q_{p_2} & \dots & q_{p_n} \end{pmatrix}$$

Ex  $S_3$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 \end{pmatrix} = (e)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 3 & 1 & 2 \end{pmatrix} = (132)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 \end{pmatrix} = (23)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix} = (123)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 3 & 2 & 1 \end{pmatrix} = (13)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 1 & 3 \end{pmatrix} = (12)$$

$$S_3 = \{ (e), (12), (13), (23), (123), (132) \}$$

$$\star (132)(123) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = (e)$$

changed the columns

$$\star (12)(13) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (123)$$

swapping column leads to wrong answer (Savrim soldan sava degil toy jecim)

|   |                      |   |                      |   |
|---|----------------------|---|----------------------|---|
| 1 | $\xrightarrow{(13)}$ | 3 | $\xrightarrow{(12)}$ | 3 |
| 2 | $\longrightarrow$    | 2 | $\longrightarrow$    | 1 |
| 3 | $\longrightarrow$    | 1 | $\longrightarrow$    | 2 |

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (132)$$